MILITARY SPECIFICATION

TEST SET, SIMULATOR TTU-291/E

This specification is mandatory for use by USAF.

1. SCOPE

1.1 This specification covers one type of simulator test set designated as the TTU-291/E hereafter referred to as the test set.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-I-8500	Interchangeability and Replaceability, Physical, of Component Parts for Aircraft
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery Of

STANDARDS

Military

MIL-STD-16	Electric and Electronic Reference Designations
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation Of

DRAWINGS

69D13070	Test Set, Simulator TTU-291/E

(Copies of the above documents which are required by contractors in connection with specific procurement functions should be obtained from the Administrative Contracting Officer or as directed by the Procuring Activity.)

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3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 Design and Construction. The test set shall be designed and constructed in conformance to Drawing No. 69D13070 and subsidiary drawings.

3.2.1 Interchangeability. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable the criteria of MIL-D-1000 shall apply and new drawings shall be prepared as required. Physical interchangeability shall be in accordance with MIL-I-8500.

3.3 Performance. Satisfactory performance of the test set shall consist of the capability to produce proper electrical simulator of a Permissive Action Link (PAL) controller.

3.3.1 Reliability. The test set shall have a Mean Time Before Failure (MTBF) rating of 350 cycles of operation under service conditions.

3.4 Environmental Requirements. The test set shall be capable of satisfactory performance after subjection to the following environmental conditions:

a. Low Temperature. The unit shall be capable of proper performance after prolonged exposure at temperature from -65°F to +165°F.

b. Vibration. The test set shall be capable of satisfactory operation after vibration, as specified in vibration tests of this specification.

c. Humidity such as encountered in tropical areas.

d. Shock incident to transportation, handling and service conditions.

3.5 Identification of Product. The test set, parts and assemblies shall be marked for identification in accordance with MIL-STD-130.

3.6 Government Loaned Equipment. When so stated in the contract, the following items will be furnished the contractor, upon his request, as government loaned property for the purpose of testing the test set:

QUANTITY	PART NUMBER	NAME OF ITEM
1	66D13260	Simulator, Switch Box, Bomb and Rocket Dispenser SMU-64/A
1	66D13250	Simulator, Selector, Bomb-Rocket Dispenser SMU-75/A

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3.7 WORKMANSHIP. The test set shall be fabricated and finished in a manner such that the criteria of appearance, fit and adherence to specified tolerances shall be observed. These criteria shall apply to all non-functioning parts as to functioning parts. Particular attention shall be given to marking of assemblies, plating, machine screw assemblage, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests. Inspection and testing of the test set shall be classified as follows:

a. Preproduction Tests

b. Acceptance Tests

4.2 Preproduction Testing

4.2.1 Preproduction Test Sample. The preproduction test units shall be representative of the production equipment. They shall be tested by a laboratory designated by the procuring activity, or when so stated in the contract, at the contractor's facility under the supervision of the procuring activity.

4.2.2 Preproduction Test Report. In the event that preproduction testing is performed at the contractor's facility, the contractor shall, at the completion of preproduction tests, prepare a preproduction test report. The preproduction test report shall be prepared in accordance with MIL-STD-831, and three complete copies shall be furnished to the procuring activity.

4.2.3 Preproduction Tests. Preproduction tests shall consist of all tests described in "4.5 Test Methods" of this specification.

4.3 Acceptance Tests. Acceptance tests shall consist of the following:

a. Individual Tests

b. Sampling Tests

4.3.1 Individual Tests. Each simulator shall be subjected to the following individual tests:

a. Examination of Product

b. Conformance to Drawings

c. Functional Tests

4.3.2 Sampling Tests. Sampling tests shall consist of the following and shall be performed in the order shown:

a. Low Temperature

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b. High Temperature

c. Vibration Test

4.3.2.1 Sampling Plan. Test sets to be subjected to the sampling tests shall be selected at random from production items as follows:

a. First 100 produced, 1 sample from each lot of 20.

b. Each additional 1000 produced, 1 sample from each lot of 100.

4.3.2.2 Rejection and Retest. When a sample item fails to satisfy the specified tests, no item still on hand or later produced shall be accepted until the extent and cause of failure is determined and resolved.

4.3.2.3 Individual Tests May Continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand or produced later shall not be made until it is determined that the items meet all requirements of this specification.

4.3.3 Defects in Items Already Accepted. The investigation of a test failure could indicate that defects exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test Conditions. Unless otherwise specified, tests shall be performed at room atmospheric conditions, and the unit shall be at room temperature when each test is started.

4.5 Test Methods

4.5.1 Examination of Product. Each test set shall be examined visually to determine compliance with all requirements of this specification for which tests are not specified. Failure to meet these requirements shall be cause for rejection.

4.5.2 Conformance to Drawings. Each test set shall be inspected to determine conformance to the engineering drawings, subsidiary drawings, and all specifications and standards referenced therein. Failure to meet these requirements shall be cause for rejection.

4.5.3 Functional Tests. Functional tests shall be performed on each test set in accordance with Table 1 to ensure that the proper electrical responses are indicated during the tests.

4.5.4 Environmental Tests. After subjection to the environmental tests, as specified in the test, the test set shall be subjected to the functional tests. Failure to satisfy the requirements of the functional tests shall be considered as failure to pass the environmental test and cause for rejection.

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TABLE I

TTU-291/E CHECKOUT PROCEDURE

1. Connect SMU-64/A and SMU-75/A Simulators
2. Connect 28V to Test Set

SMU-75/A Switches		Off, S, G,A	Off fail, en, fail	Rockets off Bombs	en	Safe	Warn	Arm	N.P.	
G, G/A A	Old New	S2	S4	S3	DS1	DS2	DS3	DS4	DS5	Comments
G	old	S	off	Bombs	---	on			on	DS3 & DS4 could flash on but after 10 sec off DS2 & DS5 should be on
G	old	G	off	Bombs	---	on 5 to 10 sec	on 5 to 10 sec	off 5 to 10 sec	on	after indication press reset and change S3 to off
G	old	S	off	Bombs	---	on	---	---	on	S3 for 1 sec then off
A	old	G	off	Bombs		on	---	---	on	No change should occur S3 on for 5 sec then off
A	old	A	off	Bombs	---	on then off	on then off	off then on	on	after indication push reset and set S3 to off
A	old	S	off	Bombs	---	on	---	---	on	S3 on for 1 sec then off
G/A	old	G	off	Bombs	---	on then off	on then off	off then on	on	reset after indication and set S3 to off
G/A	old	S	off	Bombs	---	on	off	off	on	S3 for 1 sec
G/A	new	A	off	Bombs	---	on then off	flash-ing	off then on	on	reset after indication then set S3 to off
G/A	new	S	en	Bombs	off then on in 10-15 sec	on then off	off	off	on then off	Push reset to start after indication set S3 to off
G/S	new	S	Fail Left	Bombs	off wait 15 sec	on then off	off	off	on then off	Push reset and wait 15 sec then set S3 to off
G/A	new	S	Fail Right	Bombs	off wait 15 sec	on then off	off	off	on then off	Push reset and wait 15 sec then set S3 to off
G/A	new	S	off	Rockets	on or off	on	off	off	on	S3 to off after 1 sec
G/A	new	off	off	Rockets	off	off	off	off	off	S3 to off after 1 sec

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4.5.4.1 High Temperature. The test set shall be subjected to a temperature of +165°F for at least three hours, returned to room temperature and functionally tested.

4.5.4.2 Low Temperature. The test set shall be subjected to a temperature of -65°F for at least three hours. The functional tests will be performed after returning to room temperature.

4.5.4.3 Humidity Test. The humidity test shall be performed in accordance with MIL-STD-810, Method 507. Limit time to 120 hours. Within one hour after the conclusion of the test, the functional tests shall be performed.

4.5.4.4 Shock Test. The shock test shall be performed in accordance with MIL-STD-810, Method 516, Procedure 1, Figure 516-2, (b) and (d). At the conclusion of the test, the test set shall satisfy the requirements of the functional tests.

4.5.4.5 Vibration Test. Mount unit to vibration exciter with straps. Cycle from 10 CPS to 500 CPS at .036" DA or 5G as limits at a rate of 1 octave per minute. Cycle four times in each of 3 mutually perpendicular planes. Operate unit upon completion of test.

4.5.5 Reliability Test. The test set shall be subjected to a reliability test consisting of performance of the functional test repeated 350 times. Each performance of the functional test shall be considered one cycle. Conditions of this test shall be:

a. Vibration - One cycle 20 to 2000 to 20 hz using 0.036 inch double amplitude or 5g limits (the lesser of the two) and sweeping at one octave/minute repeated once per hour of the test.

b. On-Off cycling - On for 3 hours and 10 minutes and off for 50 minutes of each four hour period of testing.

c. Temperature - 25°C ± 5°C.

d. Altitude - 0 to 5000 feet above mean sea level.

e. Humidity - 90 percent or less.

f. DC voltage 26 ± 3.

Failure of any type prior to 175 cycles shall be cause for rejection. Failure after 175 cycles and before 350 cycles requiring adjustment by means other than nominal controls shall be cause for rejection.

5. PREPARATION FOR DELIVERY. Preparation for the test set for delivery shall be in accordance with MIL-E-17555.

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5.1 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

TEST SET, SIMULATOR TTU-291/E

6. NOTES

6.1 Intended Use. The TTU-291/E is intended for use in bench checking the SMU-75 and SMU-64 simulators.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether preproduction tests will be performed by the procuring activity.

c. Levels of preservation, packaging and packing.

6.2.1 Preproduction Samples. It is expected that the contract or purchase order will specify that a minimum of three units representative of the production simulator will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

6.3 Production Delivery. No production test set should be delivered prior to full approval of the preproduction samples.

6.4 Preproduction Tests. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

b. Acceptance Tests. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

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